MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE BMB

This amendment forms a part of Military Specification MIL-M-28787/254(EC), dated 25 March 1982, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, and 6

TABLE I: Delete and substitute new table I as printed on pages 2, 3, and 4 of this amendment.

PAGE 7

4.5.1, eighth line: Delete "100°C" and substitute "85°C".

PAGE 8

6.4, second line: Delete "100°C" and substitute "85°C".

PAGE 20

FIGURE 6: Delete and substitute new figure 6 as printed on page 5 of this amendment.

PAGE 21

FIGURE 7: Delete and substitute new figure 7 as printed on page 6 of this amendment.

PAGE 23

TABLE VIII, Test condition F37, B to pin column: Add "34" between 33 and 35.

PAGE 24

FIGURE 8: Delete and substitute new figure 8 as printed on page 7 of this amendment.

PAGE 26

FIGURE 9: Delete and substitute new figure 9 as printed on page 8 of this amendment.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N478)

FSC 5963

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B32	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C32	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D4	Static output voltage, high level (I_{OH} = 0)	VI	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
E1-E4	Static output voltage, low level (I_{OL} = 0)	VII	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F36	Static output voltage, high level	VIII	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
F37-F40	Static output voltage, low level	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G4	Static output voltage, high level (V_{DD} = 15 V)	IX	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
G5-G8	Static output voltage, low level (V_{DD} = 15 V)	IX	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
H1-H32	Input current, high level	X	X	0	200	---	---	0	200	---	---	0	200	nA
J1-J32	Input current, low level	XI	X	0	-200	---	---	0	-200	---	---	0	-200	nA
K1	Power supply current, low level	XII	X	0	8	0	8	0	75	0	8	0	127	µA
K2	Power supply current, high level	XII	X	0	8	0	8	0	75	0	8	0	127	µA
L1-L4	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XIII	S	20	200	20	200	22	220	15	152	25	235	ns

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L5-L8	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XIII	Q	65	680	65	680	75	750	50	517	80	802	ns
M1-M4	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XIV	S	20	200	20	200	22	220	15	152	25	235	ns
M5-M8	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XIV	Q	65	680	65	680	75	750	50	517	80	802	ns
N1-N4	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XV	Q	10	100	10	100	11	110	7	75	12	120	ns
N5-N8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XV	Q	10	100	10	100	11	110	7	75	12	120	ns
P1-P4	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XVI	Q	8	80	8	80	9	90	6	60	10	96	ns
P5-P8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XVI	Q	8	80	8	80	9	90	6	60	10	96	ns
Q1-Q32	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XVII	S	50	500	50	500	55	555	38	380	59	592	ns
Q33-Q64	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	70	700	70	700	77	775	53	530	82	827	ns
R1-R32	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XVIII	S	65	650	65	650	72	720	49	495	77	770	ns
R33-R64	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XVIII	Q	85	850	85	850	94	940	64	645	100	1002	ns
S1-S32	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XIX	Q	20	200	20	200	22	220	15	150	24	235	ns
T1-T32	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XX	Q	26	260	26	260	29	290	19	195	32	309	ns

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U1-U32	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXI	Q	15	150	15	150	16	165	11	115	18	177	ns
V1-V32	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXII	Q	20	200	20	200	22	220	15	150	24	235	ns
W	Power supply filter capacitance	XXIII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
X1-X32	Input capacitance	XXIV	Q	2	15	2	15	2	15	2	15	2	15	pF

NOTES:

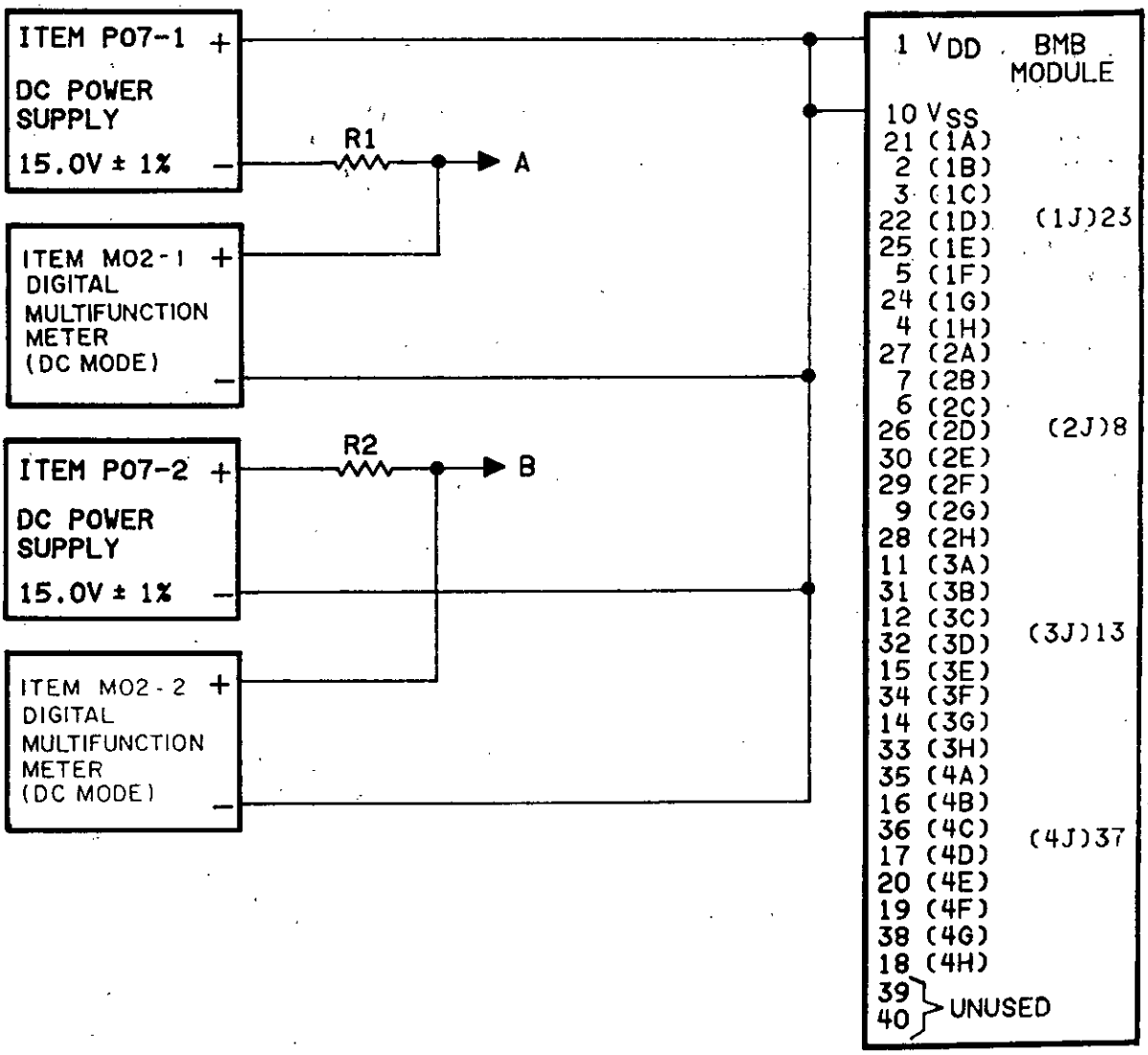
1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater)

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

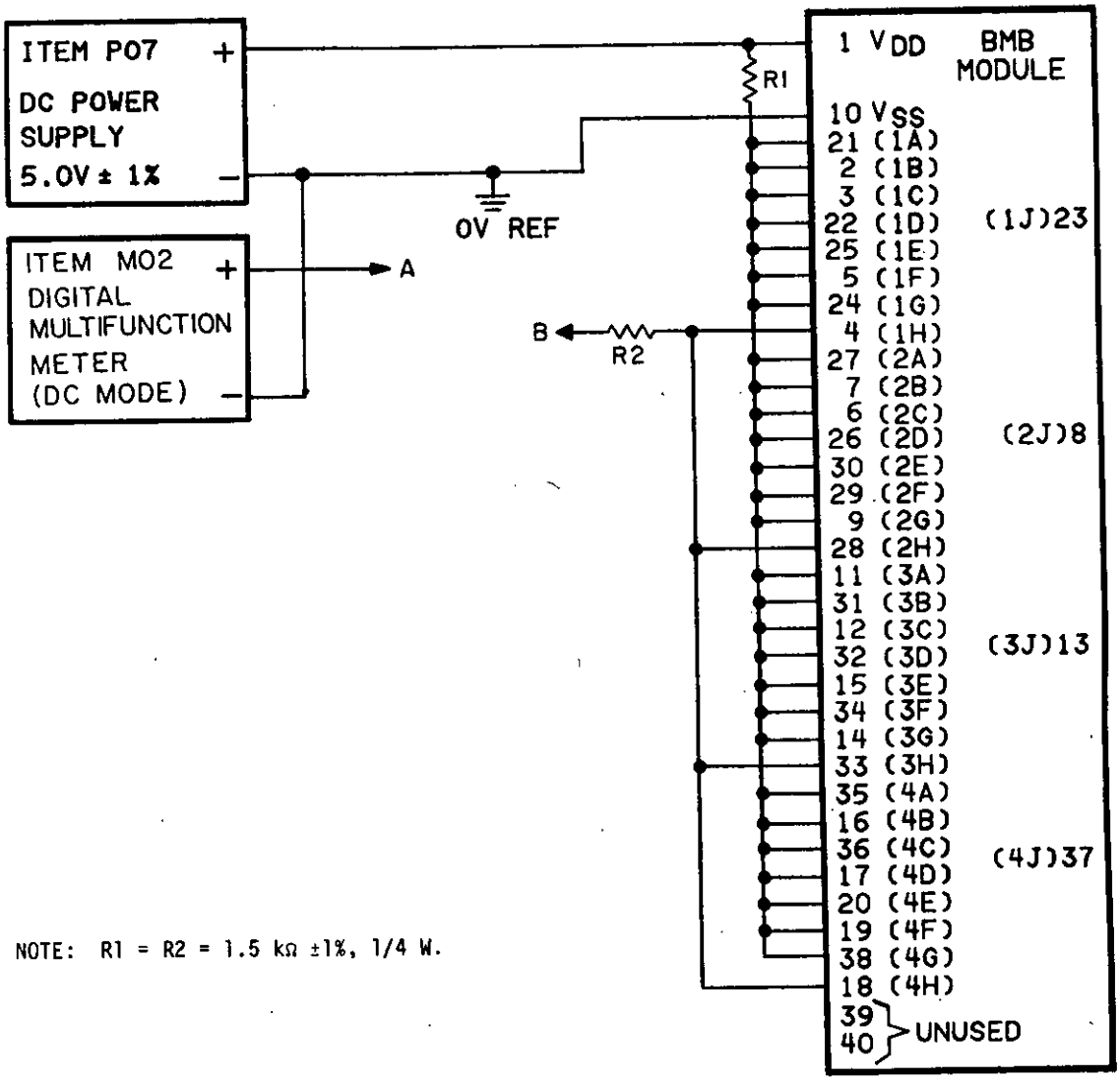
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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items P07-1 and P07-2 and M02-1 and M02-2 are not used simultaneously, only one item P07 and one item M02 is required.

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

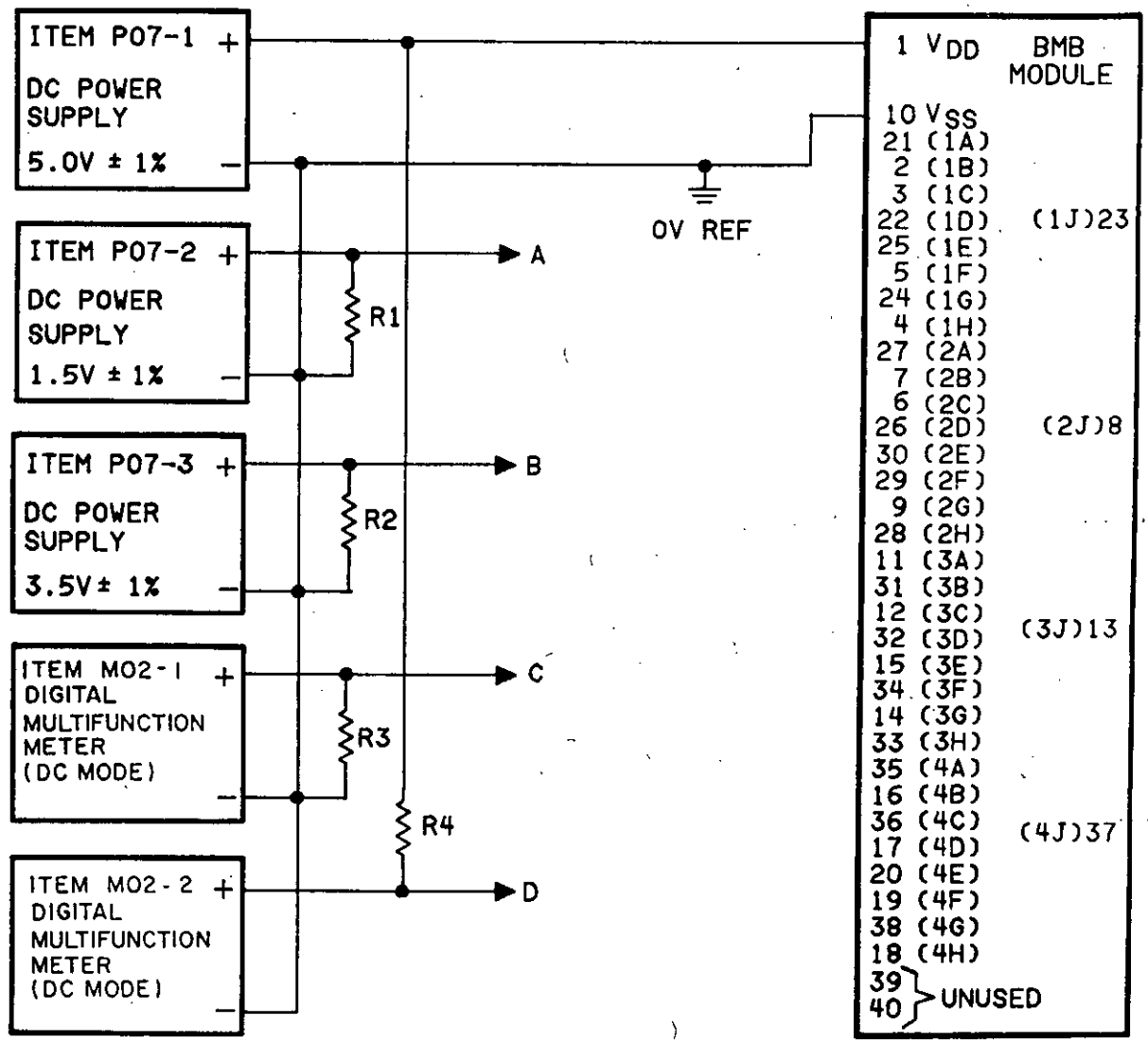
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NOTE: R1 = R2 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 7. Static output voltage (high-level, I_{OH} = 0 and low-level, I_{OL} = 0) test circuit for tables VI and VII.

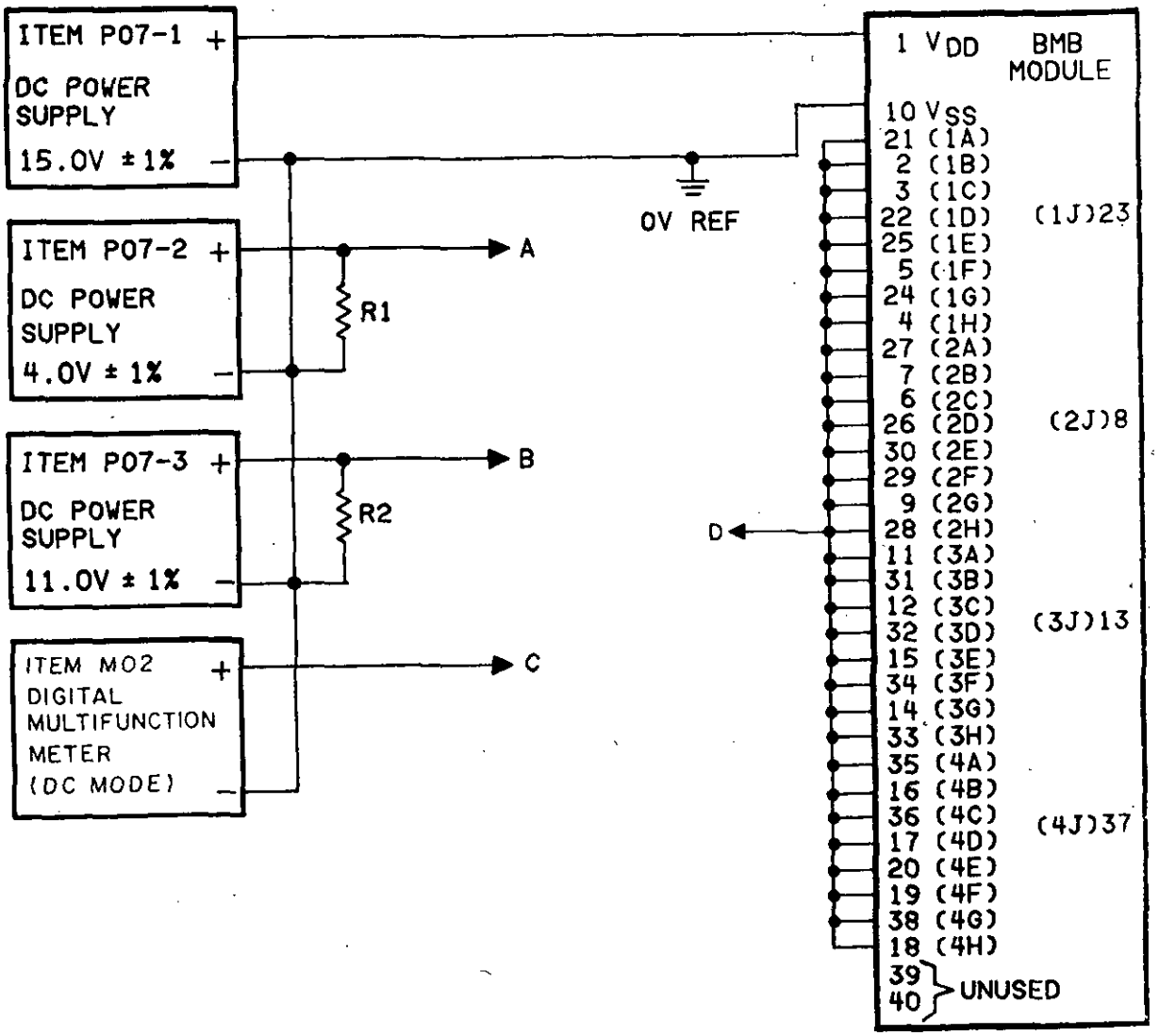
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = 18.2 kΩ ±1%, 1/4 W @ -55°C.
   = 23.2 kΩ ±1%, 1/4 W @ 0° and 25°C.
   = 27.4 kΩ ±1%, 1/4 W @ 60°C.
   = 32.4 kΩ ±1%, 1/4 W @ 85°C.
3. R4 = 7.15 kΩ ±1%, 1/4 W @ -55°C.
   = 9.09 kΩ ±1%, 1/4 W @ 0° and 25°C.
   = 10.5 kΩ ±1%, 1/4 W @ 60°C.
   = 12.7 kΩ ±1%, 1/4 W @ 85°C.

FIGURE 8. Static output voltage (high-level and low-level) test circuit for table VIII.

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NOTES:
1. R1 = 51.1 Ω ±5%, 1/2 W.
2. R2 = 576 Ω ±5%, 1/4 W.

FIGURE 9. Static output voltage (high-level and low-level, V_{DD} = 15 V) test circuit for table IX.

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